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Research Paper

Effect of Storage Time and Temperature on the Recovery of Peanut, Milk, and Gluten Residue from Environmental Swabs

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ABSTRACT

Food companies utilize environmental swabs in production facilities to validate cleaning procedures and ensure food products and food contact surfaces contain no detectable residues of unintentional food allergens. When swabs are sent to third−party laboratories as part of cleaning validation trials, the current recommendation is to ship swabs on ice with same-day shipping. However, with unforeseen delays, the temperature and duration of storage exert unknown effects on the recovery of allergen residues from swabs. The recovery of three allergenic food residues (peanut, milk, gluten) from Neogen Environmental Swabs following storage at specified temperatures and times was evaluated. Extracts of 25, 50, and 100 ppm (mg/L) peanut flour, nonfat dry milk (NFDM), and gluten (an additional 10 ppm (mg/L) spike was included for gluten) were prepared and spiked onto the swabs that were then stored at room temperature, 37 °C, 4 °C, and −20 °C for 0, 1, 3, 5, 7, 10, and 14 days. Subsequently, swabs were tested using two commercial ELISA methods (Neogen Veratox® and Morinaga) for each allergenic food. Higher and more stable recovery was obtained throughout the 14-day period across all three allergenic foods when tested using the Morinaga kits by comparison to the Veratox kits. For peanut and milk recovery from swabs analyzed using the Veratox kit, the greatest decrease in recovery was observed from day 0 to day 1. Gluten−spiked swabs tested with the Veratox kit did not demonstrate significant variability in recovery over time. Overall, higher recoveries were observed when swabs were stored at lower temperatures. These results indicate that swabs should be transported and stored at 4 °C or −20 °C until analysis. However, these results are limited to the Neogen Environmental Swabs and the ELISA kits evaluated in the current study. Further evaluation of additional protein targets, ELISA kits, and swab types is warranted to determine if these results are consistent for alternate targets, extractions, and swab types.

In the U.S.A., food allergies affect approximately 5% of adults and 8% of children having one or more food allergies (Gupta et al., 2018, Feng et al., 2023). Due to this high prevalence, food allergies are of great concern to both food-allergic consumers and the food industry. Over the years, rules and regulations have been established to ensure food companies are supplying accurately labeled products to consumers by monitoring and controlling the risk of food allergens in their facilities. In the U.S.A., federal regulations require the source labeling of ingredients derived from nine major allergenic foods including cows' milk, crustacean shellfish, egg, fish, peanut, soybean, tree nuts, wheat, and sesame seed (FDA, 2004; FDA, 2024). In addition, the Food Safety Modernization Act (FSMA) of 2011 highlights that good manufacturing practices (GMPs) should be in place to minimize unintentional cross-contact of food allergens (FDA, 2011). Despite these regulations, undeclared food allergens were responsible for between 30 and 50% of packaged food product recalls from 2010 to 2019 (FDA, 2019).

To test for the presence of allergens and the cleanliness of shared equipment, environmental swabs may be used by the food industry. These swabs can be coupled with lateral flow devices (LFDs), which offer quick, in-house indications of the presence of detectable allergen residues on equipment surfaces (Baumert & Jayasena, 2025). However, in some cases, further assessment of the swabs is desired by independent, third-party laboratories using microwell-based, quantitative allergen-specific enzyme-linked immunosorbent assay (ELISA) formats. These swabs can be used to sample different types of surfaces, valves, or other hard-to-reach/clean areas that could represent allergen harborage areas to obtain representative samples for ELISA analysis (Flanagan, 2025).

When swabs are sent to third-party laboratories, the current recommendation is that swabs be sent via same-day shipping. Some

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uncertainty exists regarding the reliability of the results, as they may be affected by the duration of time that has passed between the time of swabbing and the time of testing for the presence of allergenic residues, as well as the conditions, including temperature of storage or shipment, of the swabs. To date, little research has been conducted investigating the recovery of allergen residues from environmental swabs following storage under different conditions.

Several studies have explored the stability of direct protein extracts subjected to different storage conditions. One study showed that peanut flour extracts remain stable when stored at 2–8 °C for 12 months (Berglund et al., 2017). The stability of these protein extracts indicates that proteins are more stable at lower temperatures. Similarly, a study looking at the stability of cereal allergens found that protein extracts from wheat, rye, and barley were stable for up to 21 months when stored at 4 °C in a 50% glycerol solution (Varjonen et al., 1996). However, storage in glycerol is not possible with the use of environmental swabs used by the food industry. A more recent study, however, explored the recovery of milk and gliadin from different swab types following storage. Minimal differences in the recovery of gliadin and milk residues from various types of swabs occurred between day 1 and subsequent days of storage, although variable results were obtained among the different swab types (Barrere et al., 2020). Thus, while swab type appears to influence allergen recovery, there has yet to be a systematic study examining the recovery of different allergens from a single swab type directly spiked with known amounts of allergenic material as a function of storage temperatures and times.

To fill this gap, the recovery of three major allergenic foods (peanut, milk, and wheat gluten) from a particular type of environmental swab under differing conditions of storage temperature and time was evaluated in the current study. Consequently, this study was designed to evaluate the effect that different storage conditions have on the recovery of allergen residues from environmental swabs. By ensuring that each swab contains the same concentration of allergenic material, the direct impact of storage time and temperature could be measured. In this study, allergen recovery from a specific type of environmental swab having a known amount of peanut flour, nonfat dry milk (NFDM), or gluten in the form of an extract was evaluated when stored for a predetermined number of days (0, 1, 3, 5, 7, 10, and 14 days) at four different temperatures (room temperature, 37 °C, 4 °C, and −20 °C).

Materials and methods

Environmental swabs. Allergen Environmental Swabs (Catalog No. 8432S) were purchased from Neogen Corporation (Lansing, MI).

Spike material. Partially defatted light roast peanut flour containing 12% fat and 50% protein was purchased from Golden Peanut and Tree Nuts (Alpharetta, GA). Carnation® nonfat dry milk (NFDM) and Gold Medal™ All-Natural Whole Wheat Flour were purchased from a local grocery store in Lincoln, NE.

Commercial ELISA kits. Six commercial ELISA kits were used in this study: Veratox® for Peanut Allergen (Product No. 8430), Veratox® for Total Milk Allergen (Product No. 8470), and Veratox® for Gliadin R5 (Product No. 8510) ELISA kits from Neogen Corporation (Lansing, MI); Peanut ELISA kit (Catalog No. M2104), Casein ELISA kit II (Catalog No. M2113), and Wheat/Gluten (Gliadin) ELISA kit II (Catalog No. M2114) from Morinaga Institute of Biological Science, Inc. Japan.

Preparation, inoculation, and analysis of spiked swabs. Initial 5000 ppm (mg/L) extracts of peanut flour and NFDM were prepared separately. The peanut and NFDM extracts were prepared by mixing 0.1 g of peanut flour or NFDM in 20 mL of the preheated (60 °C) extraction buffer provided by the respective kit manufacturers and extracting at 60 °C in a shaking water bath for 20 min.

ELISA results for the wheat flour spikes were quantified in terms of ppm (mg/L) gluten. For analysis with the Veratox Gliadin R5 ELISA kit, a 800 ppm (mg/L) gluten extract was prepared by mixing 0.1 g of whole wheat flour with 5 mL of the Veratox Cocktail Solution and extracting at 50 °C in a shaking water bath for 40 min. The sample extracts were cooled to room temperature (RT), and 15 mL of 80% ethanol was added. The resulting extracts were shaken at RT for 1 h. For gluten samples tested using the Morinaga Wheat/Gluten (Gliadin) ELISA kit, a 5000 ppm (mg/L) wheat flour extract was prepared by mixing 0.1 g of whole wheat flour with 20 mL of the Morinaga extraction buffer, followed by extracting in a shaking water bath at 60 °C for 20 min. Thus, for both the Veratox and Morinaga ELISA methods, 0.1 g of whole wheat flour was extracted in a total volume of 20 mL of the respective buffers. Assuming 100% solubility, the concentration of the resulting extract is 5000 ppm (mg/L) wheat flour. Whole wheat flour contains approximately 16% gluten proteins. Thus, an extract of 5000 ppm (mg/L) whole wheat flour is equivalent to an extract of 800 ppm (mg/L) gluten.

The resulting extracts for all three allergenic foods were centrifuged at 4200 rpm for 15 min at RT, and the supernatant was used to prepare serial dilutions of 25, 50, and 100 ppm (mg/L) peanut flour, NFDM, and gluten. An additional 10 ppm (mg/L) gluten dilution was prepared from the wheat flour extracts. 0.01 M phosphate−buffered saline (PBS), pH 7.4, was used as the diluent for peanut and NFDM extracts, and 80% ethanol was used as the diluent for wheat flour extracts. 100 μL of each of the prepared dilutions was applied directly onto the Neogen Environmental swabs. The spiked swabs were stored at four different temperatures (RT, 37 °C, 4 °C, and −20 °C) for a predetermined number of days (0, 1, 3, 5, 7, 10, and 14 days). Swabs were spiked in duplicate and tested in duplicate using the representative ELISA test kits in accordance with the instructions provided with each test kit, with minor modifications to accommodate for swab testing. In order to fit within the range of the standard curve for each kit, further dilutions of the samples were tested when necessary. Following ELISA analysis, the absorbances were read according to kit instructions using a microplate reader (BioTek Instruments Inc., USA). The protein concentrations were interpolated from a standard curve generated using the Veratox Software for the Veratox kits or GraphPad Prism 4 software for the Morinaga kits. Dilution factors were factored into the calculations where appropriate. The ppm (mg/L) results obtained with each of the ELISA methods were divided by the expected recovery, and the percentage recovery was calculated for all swab samples using the equation (recovered protein value in ppm /expected protein value in ppm) × 100. Following this, the results were normalized for day 0, and the normalized percentage recovery for each day of storage for all storage temperatures was calculated for all three allergenic foods. By normalizing the percentage recovery for day 0, the effects of storage temperature/duration on allergen recovery can be better understood.

SDS-PAGE analysis. Extracts of 5000 ppm (mg/L) peanut flour and NFDM prepared in the Veratox extraction buffer (0.01 M PBS, pH, 7.4) stored at four different temperatures (RT, 37 °C, 4 °C, and −20 °C) for a predetermined number of days (0, 1, 3, 5, 7, 10, and 14 days) were analyzed using SDS-PAGE under nonreducing conditions. For gel electrophoresis, 4–20% Mini-Protean TGX Precast Gradient Gels (Bio-Rad, USA) were used. The gel apparatus was run for 30–45 min at 200 V, and the protein bands were visualized using Coomassie Brilliant Blue G-250 staining solution (Bio-Rad, USA). Precision Plus Protein Dual Xtra Standards (Bio-Rad, USA) were used as molecular weight markers.

Statistical analysis. Statistical analyses were completed using the SAS 9.4 software package. Dunnett's test was used to determine the statistical significance of mean recovery for each storage day in comparison to recovery from day 0 for each respective concentration and at each storage temperature. The level of significance was set at $p \leq 0.05$.

Results and discussion

The Neogen Environmental swabs were spiked separately with peanut, milk, and gluten extracts at known concentrations. The spike con-

centrations evaluated in the current study were selected based on the quantification range of the selected ELISA methods. Table 1 provides the ELISA kit quantification ranges, and the concentrations of each of the three allergen extracts the swabs were spiked with in the study.

Recovery of peanut from spiked swabs under different storage conditions. Previous studies have found that allergen recovery is affected by different swab types (Galan-Malo et al., 2017, Barrere et al., 2020), but overall, allergen recovery has been found to be best when swabs designed for recovery of allergen/protein residues are used compared to general-use swabs (data not shown). Thus, Neogen Environmental swabs were used in this study as they are designed to recover food allergen residues from surfaces.

The recoveries of peanut residues from swabs spiked with 25, 50, and 100 ppm (mg/L) peanut flour extract following storage at different temperatures and times varied between the two kits used in this experiment. However, overall, higher recoveries were observed when swabs were stored at lower temperatures.

With the Veratox kit, the recovery of peanut flour extract from spiked swabs decreased from days 0 to 14, with the largest decrease occurring from day 0 to day 1 across all four storage temperatures and swab concentrations (Fig. 1). Nearly all recoveries were significantly lower ($p \leq 0.05$) compared to day 0, regardless of the storage temperature or peanut flour concentration. The highest decreases in recovery were observed for swabs stored at 37 °C, with the normalized recovery decreasing to approximately 20% of expected recovery by day 14 (Fig. 1D). Better recoveries were obtained for swabs stored at RT (Fig. 1C) compared to swabs stored at 37 °C, with recoveries between 25 and 35% by day 14. The best recoveries were obtained for swabs stored at −20 °C and 4 °C, with recoveries between 45 and 80% compared to day 0 (Fig. 1A and B).

In contrast, no significant decrease in recovery was observed from day 0 to day 1 with the Morinaga kit (Fig. 1 vs 2). A less than two-fold decrease in recovery across all four storage temperatures from days 0 to 14 was observed when swabs were tested using the Morinaga peanut kit. For swabs stored at 4 °C, RT, and 37 °C, (Fig. 2B–D), normalized recoveries remained above 60% for nearly all swab concentrations. For swabs stored at −20 °C, no significant differences in recovery of peanut residues from swabs were encountered from day 0 to day 14 (Fig. 2A), and recoveries were above 90% across all concentrations. In a study of the stability of peanut allergen extracts, Berglund et al. (2017) reported that the Ara h 1 and Ara h 2 content of peanut extracts varied by less than 5% and 1%, respectively, over 12 months of storage at 4 °C. This finding indicates that peanut extracts remain stable when stored at lower temperatures.

The observed differences in peanut recovery from stored swabs between the Morinaga and Veratox results could be due to several factors, including different extraction methods used by the two kits and/or the target proteins of each kit. Jayasena et al. (2019) reported that the recovery of peanut residues from peanut flour-incurred pastries decreased four-fold following baking with the Veratox peanut kit but showed only a two-fold decrease in recovery with the Morinaga peanut kit, demonstrating that the Morinaga kit performs better at recovering peanut from heat-processed samples. The differences in the antibody targets between the two ELISA kits could also factor into the observed differences. The Morinaga kit targets the peanut proteins Ara h 2 and Ara h 6, while the Veratox kit predominantly targets Ara h 3 (Jayasena et al., 2015). Both Ara h 2 and Ara h 6 are smaller proteins that are more heat stable compared to the larger Ara h 3 proteins which have a propensity to aggregate with heating, which affects extractability (Sen et al., 2002, Breiteneder and Mills, 2005, Flinterman et al., 2007, van Boxtel et al., 2008, de Jongh et al., 2020). Additionally, Ara h 2 is known to be quite soluble in aqueous solutions compared to Ara h 3 (Shewry et al., 1995, Liu et al., 2007). Due to their stability-especially to thermal processing, but possibly also due to their comparative solubility, Ara h 2 and Ara h 6 are considered to be more robust targets for ELISA methods (Marsh et al., 2020). Furthermore, the two ELISA kits compared in this study use different extraction buffers. The Morinaga kit used in this analysis uses 2-mercaptoethanol (2-ME), a reducing agent, and sodium dodecyl sulfate (SDS) in its extraction buffer, while the Veratox kit uses PBS as its extraction buffer. 2-ME and SDS help to resolubilize and extract proteins from samples that may have aggregated, as shown by the improved recovery of peanut proteins from light roasted peanut flour using a buffer containing 2-ME along with SDS (Khuda et al., 2015).

Aliquots of the 5000 ppm (mg/L) peanut flour extract were stored under the same conditions as the swabs, and SDS-PAGE gel electrophoresis was carried out to compare the stability of peanut proteins throughout the study period. Under nonreducing conditions, no major changes, aggregation, or degradation in the protein profiles across all four storage temperatures were observed (Fig. 3). This suggests that the stability in recovery observed with the Morinaga kit is due to either the more robust extraction methods used by the Morinaga kit or the enhanced solubility of the target proteins with the Morinaga kit.

Recovery of milk from spiked swabs under different storage conditions. The recoveries of NFDM residues from swabs spiked with 25, 50, and 100 ppm (mg/L) NFDM following storage at different temperatures and times varied between the two kits (Figs. 4 and 5). With the Morinaga kit, more stable recoveries were obtained across all stor-

Table 1
Comparison of the range of quantification of the kit standards of the six commercial kits used in the current study and the concentrations of the spiking material applied to the environmental swabs.

ELISA kit	Range of quantification of ELISA kit	Range of quantification of ELISA kit (ppm protein)	Concentrations of spike material applied to swabs	Concentrations of spike material applied to swabs (ppm protein)
Veratox® for Peanut Allergen	2.5–25 ppm total peanut	0.645–6.45[a]	25, 50, and 100 ppm peanut flour extract	12.5, 25, and 50[b]
Veratox® for Total Milk Allergen	2.5–25 ppm nonfat dry milk	0.9-9[c]	25, 50, and 100 ppm nonfat dry milk	8.75, 17.5, and 35[d]
Veratox® for Gliadin R5	2.5–40 ppm gliadin	5–80 ppm gluten[e]	10, 25, 50, and 100 ppm gluten	–
Morinaga Peanut ELISA kit	–	0.31–20 ppm	25, 50, and 100 ppm peanut flour extract	12.5, 25, and 50[b]
Morinaga Casein ELISA kit II	–	0.31–20 ppm	25, 50, and 100 ppm nonfat dry milk	8.75, 17.5, and 35[d]
Morinaga Wheat/Gluten (Gliadin) ELISA kit II	–	0.26–17 ppm gluten	10, 25, 50, and 100 ppm gluten	–

[a] The Veratox® for Peanut Allergen ELISA quantifies in ppm peanut and a 0.258 conversion factor as provided by the kit manufacturer was used to convert ppm peanut to ppm peanut protein.
[b] The partially defatted light roast peanut flour purchased from Golden Peanut and Tree Nuts (Alpharetta, GA) contains 50% protein.
[c] The Veratox® for Total Milk Allergen ELISA quantifies in ppm nonfat dry milk (NFDM) and a 0.36 conversion factor as provided by the kit manufacturer was used to convert ppm NFDM to ppm protein.
[d] NFDM purchased from a local grocery store in Lincoln, NE, contains 35% protein.

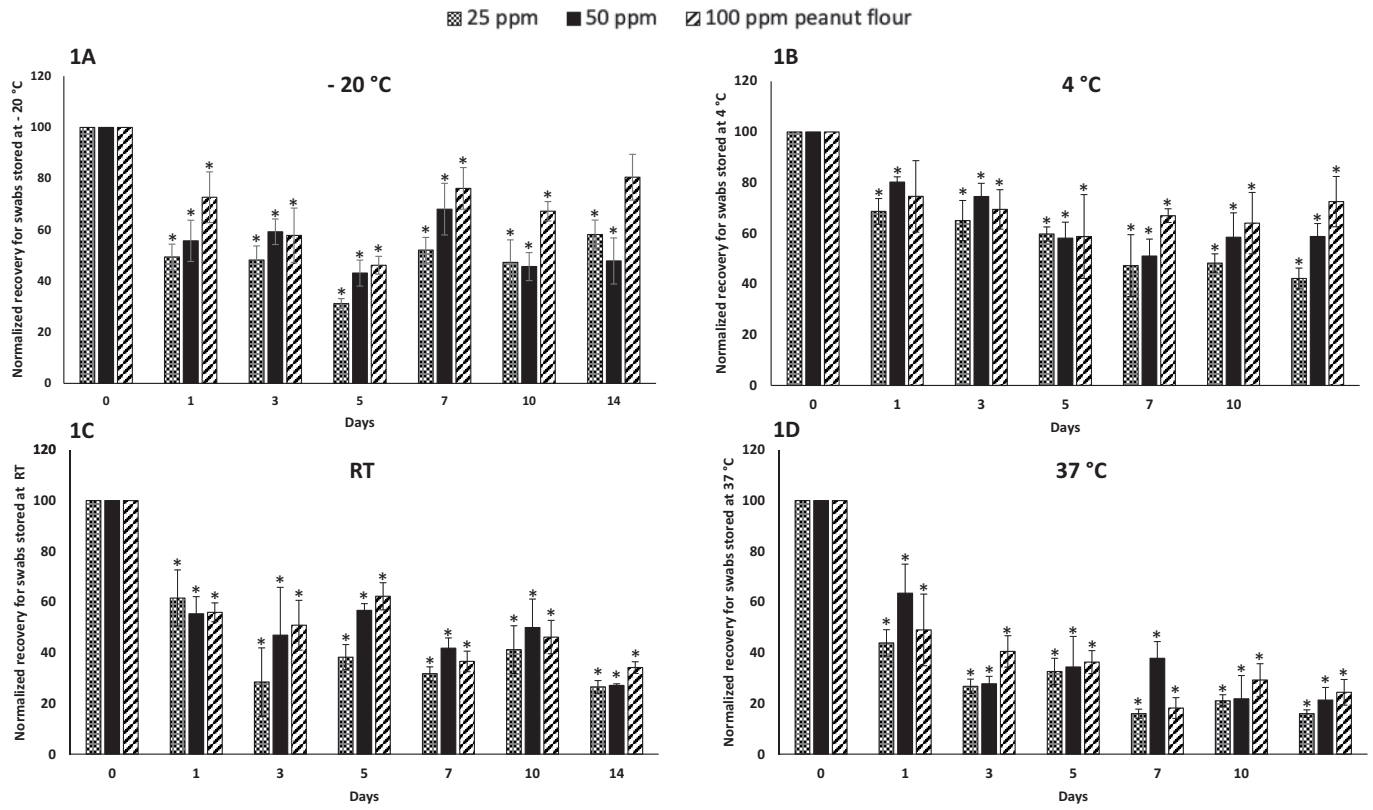


Figure 1. Normalized percentage recovery from peanut spiked swabs using the Neogen Veratox Peanut Allergen ELISA kit at (A) −20 °C, (B) 4 °C, (C) RT, and (D) 37 °C storage temperatures. Values are expressed as mean ± standard deviation (n = 2, each analyzed in duplicate). *, mean recovery for each specified day is significantly different ($p \leq 0.05$) when compared to day 0 for each respective concentration.

age temperatures and durations (Fig. 5) compared to the Veratox kit, which showed higher decreases in recovery, especially when swabs were stored at 37 °C (Fig. 4).

Similar to the Veratox peanut results, the recoveries of NFDM from swabs analyzed with the Veratox milk kit were the highest on day 0 with the largest decrease in recovery observed from day 0 to day 1 across all four storage temperatures (Fig. 4). Recovery of NFDM from swabs stored at 37 °C showed the most significant decreases compared to swabs stored at RT, 4 °C, and −20 °C across the 14-day storage period ($p \leq 0.05$). The recovery of NFDM from spiked swabs stored at RT, 4 °C, and −20 °C was 60–70% of expected recovery by day 14 (Fig. 4A–C). However, when swabs were stored at 37 °C, the normalized recovery was below 60% of the expected recovery by day 14 of storage (Fig. 4D).

For NFDM−spiked swabs tested with the Morinaga kit, less variation in recoveries over the 14 days of storage was observed. When comparing trends between the two milk kits, no sharp decrease in recovery from day 0 to day 1 was observed with the Morinaga kit (Fig. 4 vs Fig. 5). When swabs were stored at 4 °C and −20 °C, the recoveries of NFDM from spiked swabs with the Morinaga kit were comparable from day 0 across most of the days of storage (Fig. 5A and B). Slight decreases in recovery were noted for swabs stored at RT throughout the duration of storage, yet the normalized recovery remained around 80% of the expected recovery for all three spike concentrations by day 14 (Fig. 5). When 25 and 50 ppm (mg/L) NFDM spiked swabs were stored at 37 °C, the normalized recoveries were between 50 and 60%, while 100 ppm (mg/L) NFDM spiked swabs remained around 90% of expected recovery (Fig. 5D).

Overall, the recoveries of milk residues from swabs tested using the Morinaga kit were higher and more stable when compared to the Veratox kit throughout the 14 days of storage. Barrere et al. (2020) evaluated the recovery of milk and gliadin proteins using different swab types for residues swabbed from a stainless-steel surface and stored under different conditions over a 7-day period. When the swabs were tested using the RIDASCREEN® Fast Milk ELISA kit, recoveries from swabs were above the limit of detection for all swabs stored at 4 °C, even on day 7. However, when swabs were stored at 37 °C, milk was not recovered from several swab types. Overall, these results indicate that both temperature and duration of storage influence the recovery of allergenic residues from swabs and recovery is less affected when swabs are stored at lower temperatures.

Aliquots of the 5000 ppm (mg/L) NFDM extracts were stored under the same conditions as the swabs, and SDS-PAGE gel electrophoresis was conducted to compare the stability of NFDM proteins throughout the study period. In contrast to the stability of the proteins observed for peanut extracts (Fig. 3), NFDM extracts stored at RT and 37 °C and analyzed under nonreducing conditions (Fig. 6A) indicated some aggregation and degradation of milk proteins, especially from the casein proteins, from day 7 onwards when stored at RT. Similar changes were observed beginning from day 3 for extracts stored at 37 °C. Extracts stored at 4 °C and −20 °C, however, had no observable changes in the overall protein profiles over the 14 days of storage (Fig. 6B).

The Veratox Total Milk ELISA kit targets primarily κ-casein but does not detect β-lactoglobulin nor α-lactalbumin (Ivens et al., 2016). Western blotting using the conjugate polyclonal antibodies from the Morinaga Casein ELISA Kit II indicated that the antibodies from the Morinaga kit primarily target α-casein (data not shown). With both kits targeting the casein fraction of milk, the reasons for the differences in recoveries could be due to the different extraction conditions used by the kit manufacturers. As was observed with the peanut kits, the two milk kits assessed in this study differ with respect

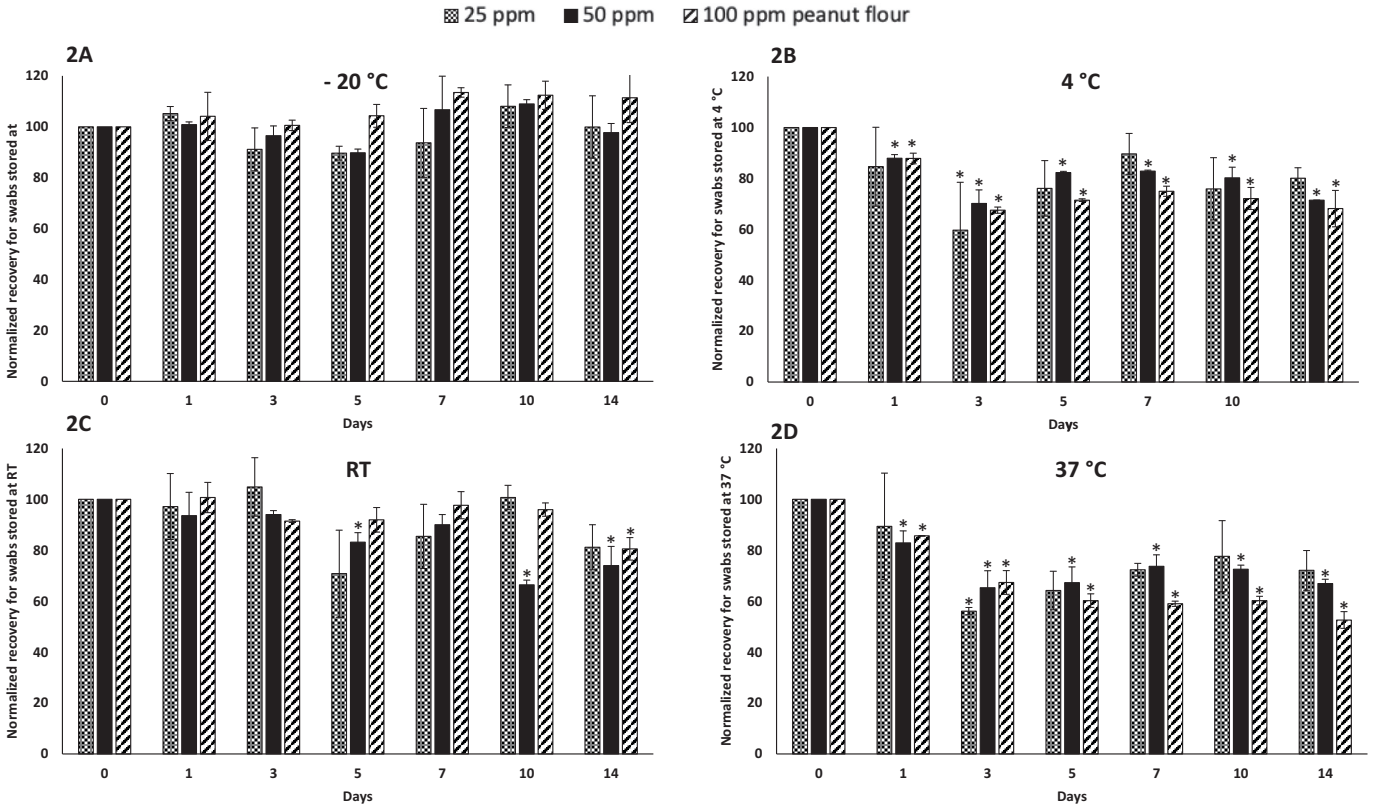


Figure 2. Normalized percentage recovery from peanut spiked swabs using the Morinaga Peanut ELISA kit at (A) −20 °C, (B) 4 °C, (C) RT, and (D) 37 °C storage temperatures. Values are expressed as mean ± standard deviation ($n = 2$, each analyzed in duplicate). *, mean recovery for each specified day is significantly different ($p \leq 0.05$) when compared to day 0 for each respective concentration.

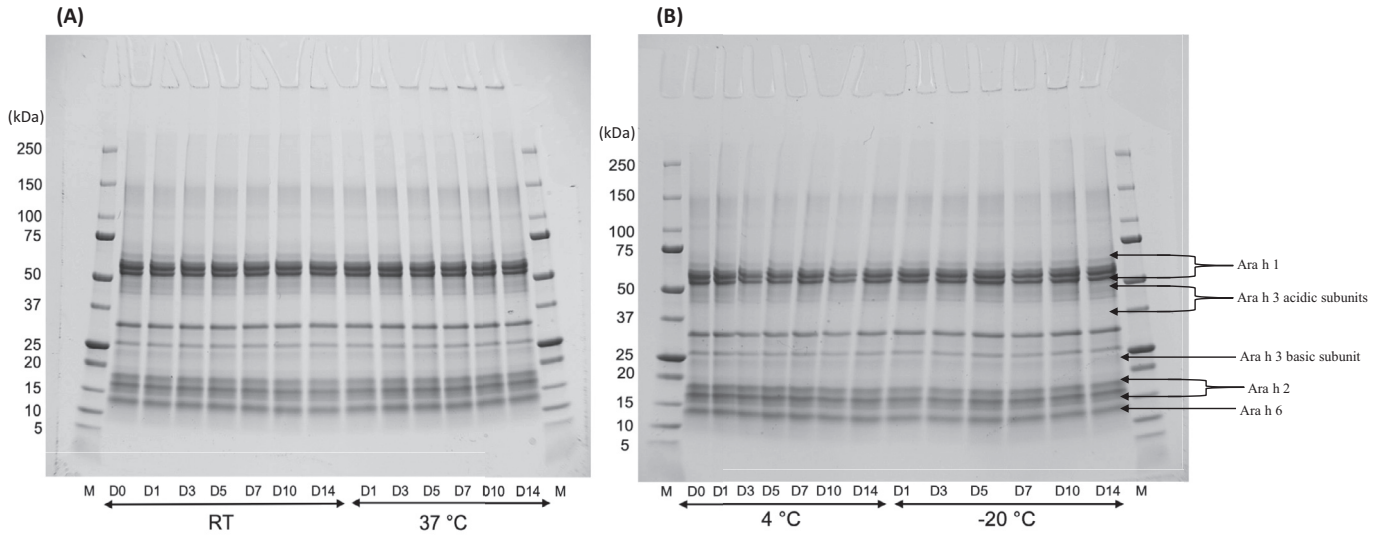


Figure 3. SDS-PAGE analysis under nonreducing conditions of 5000 ppm (mg/L) peanut flour extracts stored at (A) RT & 37 °C and (B) 4 °C & −20 °C for 0–14 days. 50 μL of sample extract was mixed with 10 μL of 6x Laemmli sample buffer, and 12 μL/lane was loaded. M; protein standard marker with MW (in kDa) for each standard indicated on the left-hand side, lanes D0 through D14 represent days 0 to 14 of storage.

to the extraction buffer type. The Morinaga kit utilizes a reducing agent, sodium sulfite, while the Veratox kit uses a PBS-based extraction method. By utilizing a more rigorous extraction method, the presence of the reducing agent in the Morinaga kit can resolubilize proteins that may have aggregated. Aggregation was clearly seen in the SDS-PAGE gels for NFDM extracts stored at RT and 37 °C. The higher recoveries observed with the Morinaga kit, regardless of the swabs' storage condition, could be explained by the use of a more effective extraction buffer.

Recovery of gluten from spiked swabs under different storage conditions. The recoveries of gluten from swabs spiked with 10, 25, 50, and 100 ppm (mg/L) gluten were evaluated using the Veratox

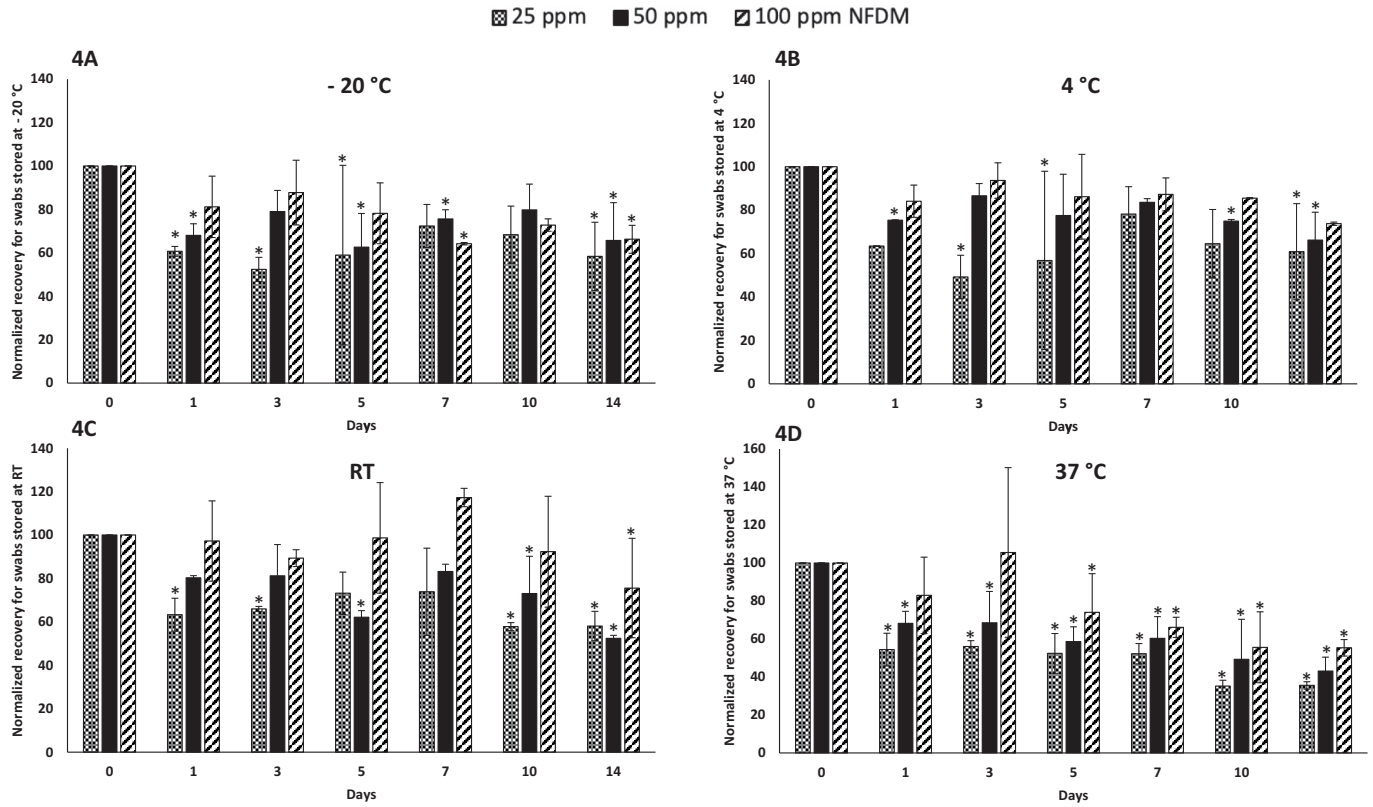


Figure 4. Normalized percentage recovery from NFDM spiked swabs using the Neogen Veratox Total Milk Allergen ELISA kit at (A) −20 °C, (B) 4 °C, (C) RT, and (D) 37 °C storage temperatures. Values are expressed as mean ± standard deviation ($n = 4$, each analyzed in duplicate). *, mean recovery for each specified day is significantly different ($p \leq 0.05$) when compared to day 0 for each respective concentration.

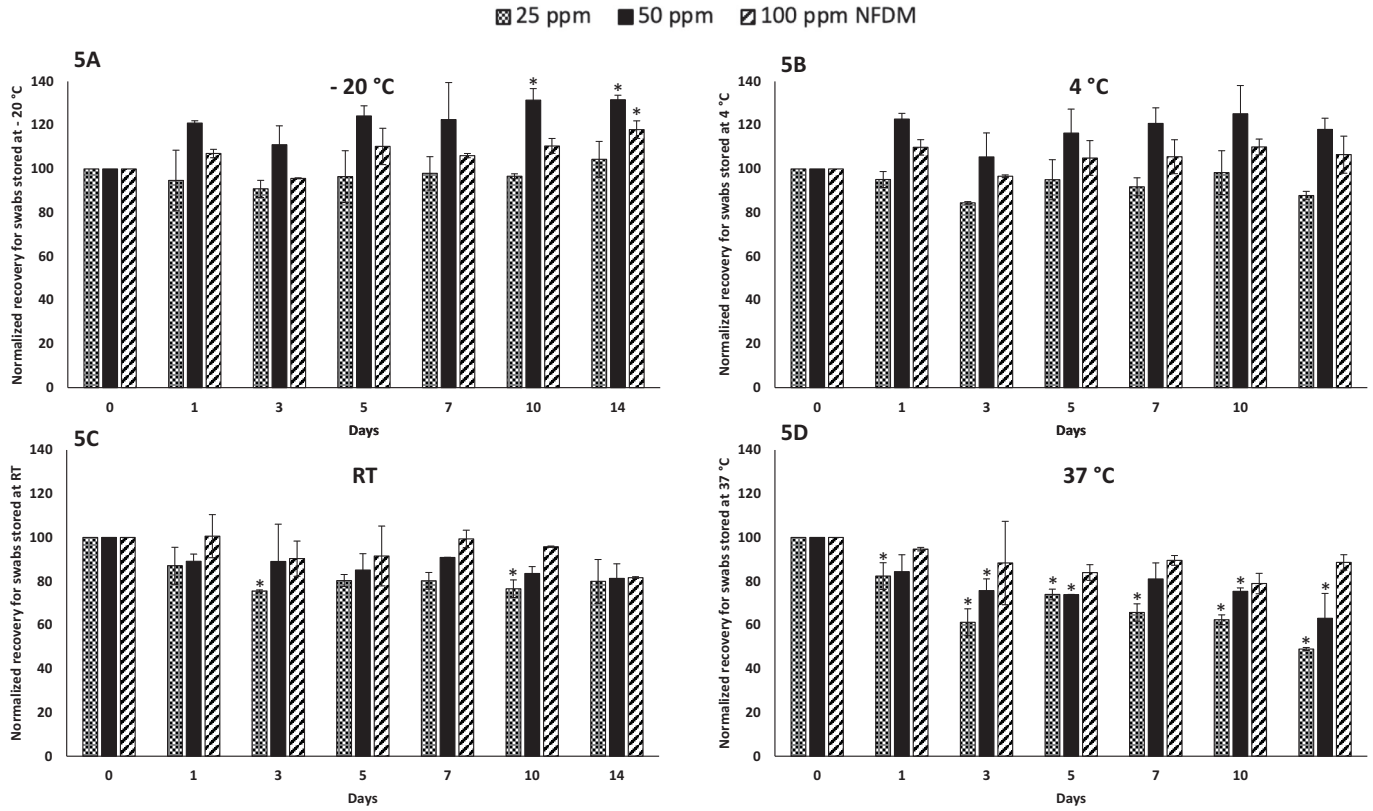


Figure 5. Normalized percentage recovery from NFDM spiked swabs using the Morinaga Casein ELISA kit II at (A) −20 °C, (B) 4 °C, (C) RT, and (D) 37 °C storage temperatures. Values are expressed as mean ± standard deviation ($n = 2$, each analyzed in duplicate). *, mean recovery for each specified day is significantly different ($p \leq 0.05$) when compared to day 0 for each respective concentration.

Figure 6

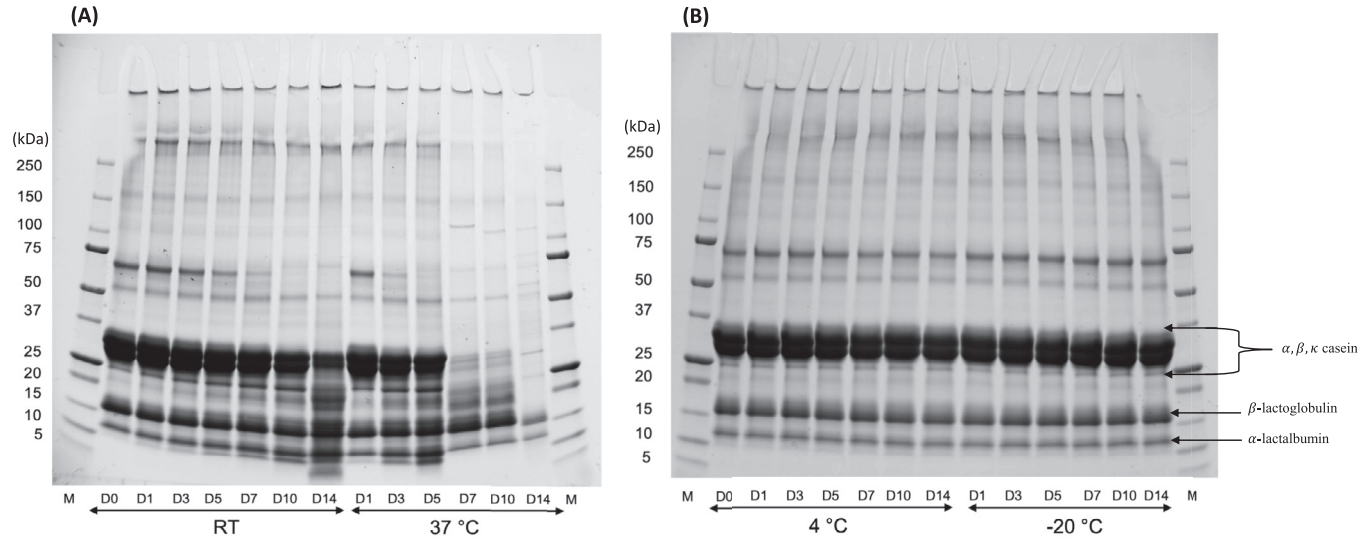


Figure 6. SDS-PAGE analysis under nonreducing conditions of 5000 ppm (mg/L) NFDM extracts stored at (A) RT & 37 ˚C and (B) 4 ˚C & −20 ˚C for 0–14 days. 50 μL of sample extract was mixed with 10 μL of 6x Laemmli sample buffer, and 12 μL/lane was loaded. M; protein standard marker with MW (in kDa) for each standard indicated on the left-hand side, lanes D0 through D14 represent days 0 to 14 of storage.

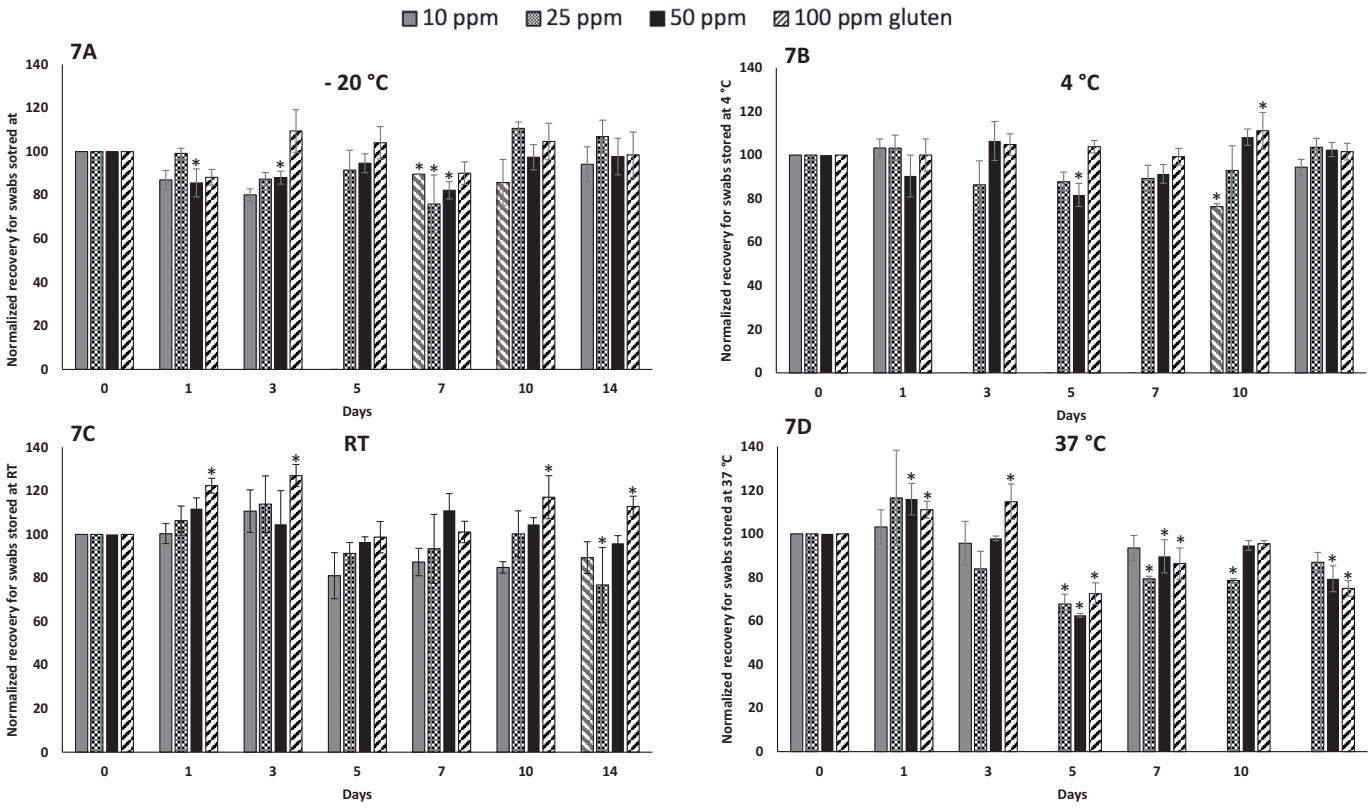


Figure 7. Normalized percentage recovery from wheat flour spiked swabs using the Neogen Veratox Gluten R5 ELISA kit at (A) −20 ˚C, (B) 4 ˚C, (C) RT, and (D) 37 ˚C storage temperatures. Values are expressed as mean ± standard deviation ($n = 2$, each analyzed in duplicate). *, mean recovery for each specified day is significantly different ($p \leq 0.05$) when compared to day 0 for each respective concentration. For 10 ppm (mg/L) gluten, checkered black bars represent all 4 (4/4) replicates were quantifiable while the solid grey bar represents at least 1/4 replicates were below the level of quantification (BLQ) of the ELISA kit.

and Morinaga commercial gluten ELISA kits following different storage conditions. Overall, the recoveries of gluten-spiked swabs were more similar between the Veratox and Morinaga gluten kits compared to the differences encountered with the Veratox and Morinaga peanut and milk kits. Gluten spiked swabs analyzed using the Veratox kit did not demonstrate a large decrease in recovery from day 0 to day 1, differing from the observations with the Veratox peanut and milk kits (Fig. 7 vs Figs. 1 and 4).

While gluten was recovered from the 10 ppm (mg/L) gluten spiked swabs on day 0 and day 1 across all four storage temperatures, at least 1/4 replicates analyzed with the Veratox kit were below the limit of quantification (BLQ) of 5 ppm (mg/L) gluten beginning from day 3 for swabs stored at 4 °C, day 5 for swabs stored at −20 °C and 37 °C, and day 14 for swabs stored at RT (Fig. 7). For swabs stored at 37 °C, the normalized recoveries remained above 60%, and when swabs were stored at RT, 4 °C, or −20 °C, normalized recoveries remained at or above 80% for 25, 50, and 100 ppm (mg/L) gluten spiked swabs.

In contrast to the Veratox Kit, quantifiable results were obtained with the 10 ppm (mg/L) gluten spiked swabs at all four storage temperatures across four replicates with the Morinaga kit. Moreover, no large decreases in recovery were observed throughout the 14-day period across all four storage temperatures (Fig. 8). In fact, the decrease in recovery was less than 20% of the expected recovery for nearly all swab concentrations at all storage temperatures, with the exception being on day 7 for 10 ppm (mg/L) spiked swabs stored at 37 °C (Fig. 8D).

Apart from some of the 10 ppm (mg/L) gluten spiked swabs tested with the Veratox kit, quantitative recoveries were obtained over the 14-day period with both gluten kits. These results are consistent with the observations reported by Barrere et al. (2020), where they evaluated the recovery of gliadin from different swab types following the swabbing of stainless steel. Using the RIDASCREEN® FAST Gliadin ELISA kit for analysis, the authors showed that with the cotton and polyurethane swab types, the recovery of gliadin was stable throughout the 7-day storage conditions, even when stored at 37 °C. The RIDASCREEN® FAST Gliadin ELISA kit and the Veratox kit both use the R5 monoclonal antibody for detection, which is known to target a heat-resistant epitope from gliadin (García et al., 2005; Sharma, 2012), while the Morinaga kit uses polyclonal antibodies that primarily target gliadin proteins (Sharma, 2012). Both the RIDASCREEN® and Veratox kits also use a cocktail solution as part of the extraction procedure, which contains 2-ME and guanidine hydrochloride as reducing and denaturing agents, respectively. These agents help break up gluten proteins, which may have aggregated over time (García et al., 2005, Wieser, 1998). While the Morinaga kit does not use the same cocktail solution, it does utilize sodium sulfite as a reducing agent to help disrupt the gluten matrix and thereby increase the efficacy of protein extraction. Thus, the use of reducing and/or denaturing agents in the extraction buffers of both kits most likely contributes to the stability that was observed with both the Veratox and Morinaga gluten kits, even when swabs were stored for extended periods of time at elevated temperatures.

In summary, this study evaluated the effect of storage temperature and duration on the recoverability of peanut, milk, and gluten residues from spiked swabs following storage at different temperatures for a predetermined number of days. Recovery was evaluated using commercial peanut, milk, and gluten ELISA kits from Neogen Corporation and Morinaga Institute of Biological Science Inc. With peanut and milk swabs, less variability and lower decreases in recovery were observed when swabs were tested with the Morinaga kits in comparison to the Veratox kits. This observed difference may be due to the use of a

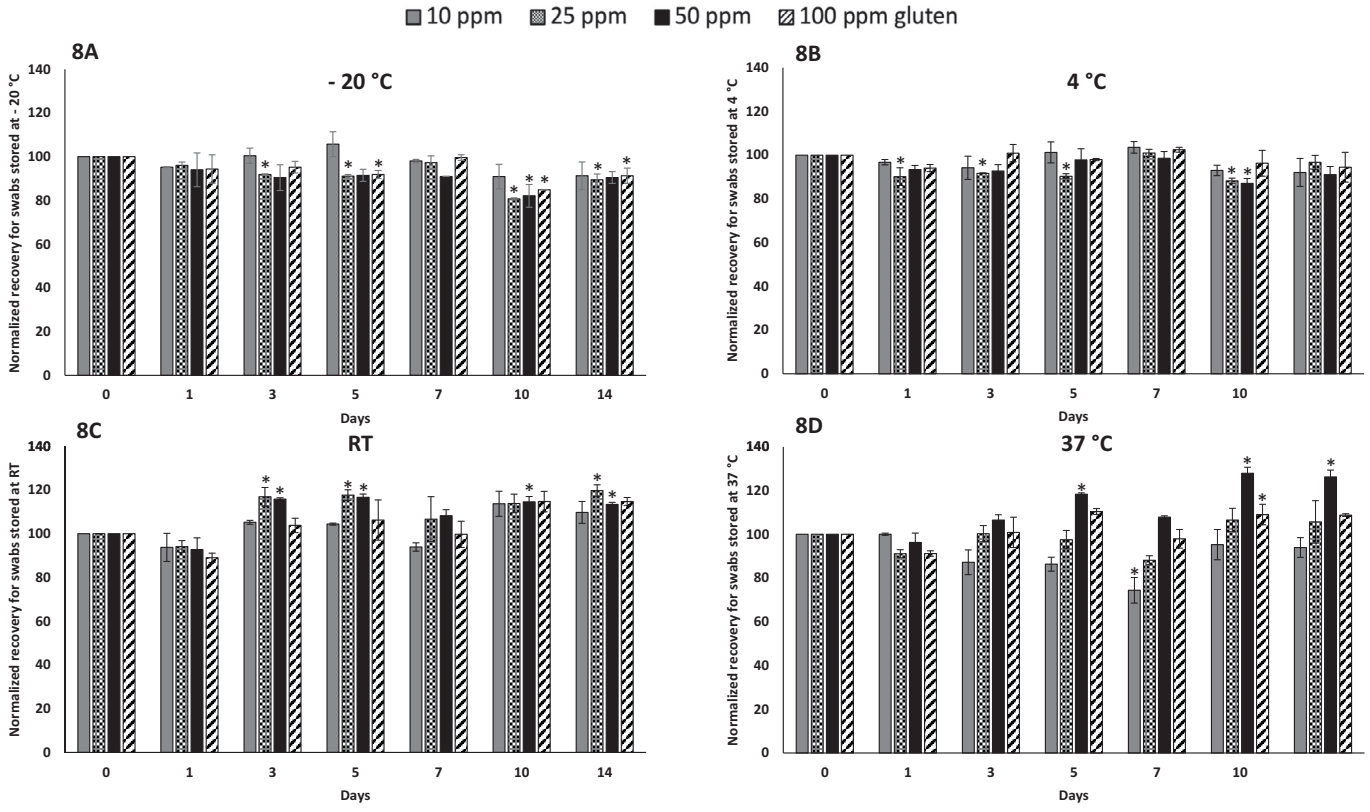


Figure 8. Normalized percentage recovery from wheat flour spiked swabs using the Morinaga Wheat/Gluten (Gliadin) ELISA kit II at (A) −20 °C, (B) 4 °C, (C) RT, and (D) 37 °C storage temperatures. Values are expressed as mean ± standard deviation (n = 2, each analyzed in duplicate). *, mean recovery for each specified day is significantly different ($p \leq 0.05$) when compared to day 0 for each respective concentration.

reducing agent in the extraction buffer of the Morinaga kit for both peanut and milk recovery. More consistent and comparable results between the two ELISA kits were obtained with the recovery of gluten, which is attributed to the similar presence of reducing agents in the extraction buffers for both kits. Overall, recoveries were more stable for the gluten−spiked swabs compared to the peanut and NFDM −spiked swabs throughout the 14 days of storage. When stored at −20 °C and 4 °C, both the Veratox and Morinaga ELISA kits had higher recoveries by comparison to swabs stored at RT and 37 °C throughout the 14-day period. These results support the recommendation that swabs be sent to testing laboratories on ice and stored at either −20 °C or 4 °C until further analysis. These results are limited to the protein targets, ELISA kits, and swab types evaluated in the current study. The antibodies used in commercial ELISA kits are raised against different protein targets and are thus heavily kit−dependent. For example, the Veratox peanut kit predominantly targets peanut Ara h 3 while the predominant protein target of the Morinaga ELISA kit is Ara h 2. Likewise, the protein target of the Veratox total milk ELISA and the Morinaga casein ELISA kit II are κ-casein and α-casein, respectively. As mentioned previously, the recovery percentage for a given ELISA method is dependent on several factors, including the thermal stability of the protein target. Additionally, ELISA methods differ from one to another depending on their sample extraction procedures, the protein standards used in their calibration curves, the range of quantification and the expression of results. Thus, the results reported among different kits can sometimes be quite variable. Consequently, further evaluation of additional protein targets, ELISA kits, and swab types is needed to determine if similar results will be observed for other food allergens.

CRediT authorship contribution statement

Jessica Humphrey: Writing – original draft, Methodology, Formal analysis, Data curation, Conceptualization. **Shyamali Jayasena:** Writing – review & editing, Supervision, Methodology, Formal analysis, Conceptualization. **Steve L. Taylor:** Writing – review & editing, Supervision, Resources, Project administration, Methodology, Funding acquisition, Conceptualization. **Joseph L. Baumert:** Writing – review & editing, Supervision, Resources, Project administration, Methodology, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary material

Supplementary material to this article can be found online at https://doi.org/10.1016/j.jfp.2025.100645.

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